TOWARDS TRANSPARENT EMBEDDINGS FOR BIOMARKER DISCOVERY

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ABSTRACT

Foundation models often yield highly predictive embeddings; however, their semantic content can be opaque, which limits real-world use. We introduce BioLENS, a framework for ordered-outcome settings that models latent structure as a smooth embedding–outcome trajectory g, then extracts function-level features (e.g., trend, curvature, extrema) to summarize how representations vary across outcome levels. In a biomarker-based minimum viable prototype (MVP), we instantiate g using embeddings of neural time series with a smoothing spline, and translate clinically motivated structure into explicit constraints to evaluate whether latent geometry aligns with meaningful outcome progression. More generally, the MVP illustrates how domain-informed desiderata can be translated into quantitative, testable constraints for comparing encoders, and it points toward implicit representation learning objectives that incorporate such constraints to encourage embeddings that are not only predictive but also outcome-relevant and interpretable. Code is available at: https://github.com/jacksampiere/biolens

1 INTRODUCTION

The interpretability challenge for latent spaces. In high-stakes domains such as finance, medicine, and law, it is often not enough for a model to perform well. Stakeholders also need principled ways to understand what drives its outputs and when those outputs should be trusted. Traditional interpretability tools such as feature importance, permutation tests, and Shapley-style attributions (Breiman, 2001; Fisher et al., 2019; Lundberg & Lee, 2017) are effective when model inputs correspond to meaningful, human-defined variables; however, when features are opaque, entangled, or lack clear semantics, attribution becomes difficult to translate into actionable explanations.

Such limitations have become increasingly salient amid the rapid proliferation of foundation models in recent years (Devlin et al., 2019; Brown et al., 2020). Foundation models are typically trained at scale to produce general-purpose latent representations (embeddings) that compactly encode increasingly rich and complex input information in a fixed-dimensional latent space (Bengio et al., 2012). In practice, these embeddings are often consumed primarily as a black-box basis for downstream fine-tuning rather than as objects of scientific interpretation (Bommasani et al., 2021).

Motivation: neurobiomarker discovery. The limited semantics of embeddings is especially consequential in neuroscience and neurotechnology, where raw measurements are high-dimensional, noisy, and mechanistically complex. Electroencephalography (EEG) measures electrical activity recorded non-invasively from scalp electrodes, often at high sampling rates and across multiple channels. This makes it a natural target for representation learning, yet a difficult substrate for interpretation (Biasiucci et al., 2019).

Recent work suggests that large-scale self-supervised pretraining can learn representations from EEG that transfer across tasks and cohorts, improving performance in clinically relevant settings (Chau et al., 2025). Yet, the limited interpretability of these representations constrains their adoption for applications that demand transparent clinical reasoning, including diagnosis, stratification, and treatment planning, even when the representations appear to capture highly relevant information. Existing attempts to understand foundation model embeddings, such as correlating embedding coordinates with handcrafted signal features or metadata, linear probing models, and

clustering-based analyses provide partial insight (Kim et al., 2018; Alain & Bengio, 2016; van der Maaten & Hinton, 2008). However, they rarely yield explanations that connect quantitative embedding structure to human-interpretable concepts.

A promising framework: TIMEVIEW. To move beyond correlational and probing-style analyses of embeddings, we draw inspiration from recent interpretability advances in time series forecasting. Specifically, Kacprzyk et al. (2024) present TIMEVIEW: a "bi-level transparency" framework that formalizes how humans naturally describe trajectories in terms of their trends (global shape) and the properties of those trends (e.g., minima, maxima), rather than as collections of pointwise values. Motivated by this top-down view of trajectory comprehension, interpretability is framed as understanding how perturbations of the input change the overall trend of the predicted trajectory (level 1) and the key properties of that trend (level 2). This approach contrasts bottom-up feature attribution approaches that operate over individual prediction horizons or pre-specified time points.

Contributions. Our contributions are as follows:

1. We introduce BioLENS, a framework for interpreting embeddings in ordered-outcome settings motivated by biomarker discovery.

2. BioLENS yields human-interpretable summaries of how representations change across outcome levels and uses them to derive testable, domain-informed criteria for candidate embeddings.

3. We instantiate BioLENS in an EEG-based minimum viable prototype (MVP), illustrating how domain expectations can guide both embedding evaluation and representation learning objectives beyond downstream performance.

2 BioLENS

BioLENS (Biomarker-discovery-driven Latent Embedding iNterpretation via Splines) is a TIMEVIEW-inspired framework for interpreting embeddings by encoding structured relationships between a latent space and an ordered outcome. Given embedding–outcome pairs $\{(z_i, y_i)\}$, where $z_i \in \mathbb{R}^k$ is a k-dimensional embedding vector and y_i is ordered (e.g., an ordinal label, risk score, or outcome quantile), we model the embedding as a continuous trajectory g indexed by y. We estimate g with a smooth fit, which makes it straightforward to extract derivative-based, human-interpretable structure.

Analogous to TIMEVIEW's decomposition of trajectories into locally consistent motifs separated by transition points, BioLENS uses the geometry of g to produce interpretable statements about trend direction, curvature, and qualitatively meaningful changes as the outcome varies. This yields a representation-level analogue of TIMEVIEW's composition map: a mapping from outcome intervals and corresponding embeddings to descriptions of how the representation evolves.

Crucially, BioLENS is not only post hoc: it provides a way to translate domain expectations (e.g., "risk should rise sharply over a clinically relevant interval, then saturate") into explicit constraints on g (e.g., positive slope over an interval, concave-down curvature, transition points near clinically meaningful bounds). These constraints can be used to evaluate whether an embedding supports a target use case and to define objectives/regularizers for learning embeddings that exhibit desired structure beyond downstream task performance.

Figure 1 summarizes BioLENS end-to-end. The top row depicts the generic pipeline (Section 3), while the shaded bottom row shows the MVP instantiation (Section 4).

3 Formulation

We first formulate the problem generically. Section 4 then outlines an EEG-based MVP.

3.1 Setup and Assumptions

Let $X = \{x_1, x_2, \ldots, x_n\}$ denote inputs and let $E : X \to \mathbb{R}^k$ be a fixed encoder that maps each x_i to an embedding $z_i = E(x_i) \in \mathbb{R}^k$. Let $Y = \{y_1, y_2, \ldots, y_n\}$ be the corresponding outcomes. We assume outcomes lie on a one-dimensional ordered domain $I \subset R$, with $y_i \in I$.

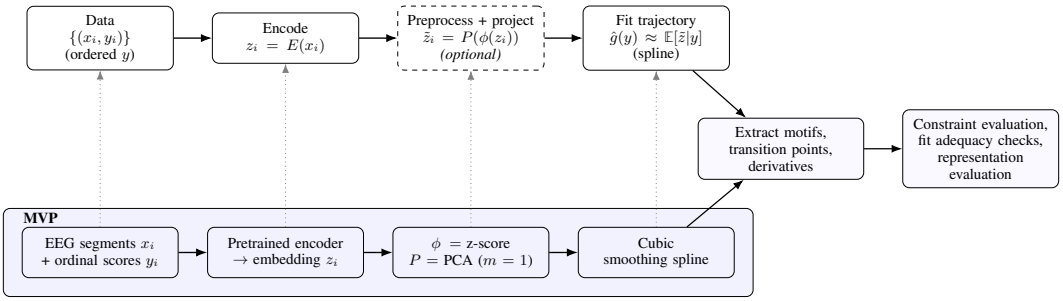

Figure 1: BioLENS algorithm flow: generic pipeline (top) with an MVP instantiation (bottom).

3.2 Optional Preprocessing and Projection

Optionally, we apply a preprocessing map $\phi : \mathbb{R}^k \to \mathbb{R}^k$ (e.g., normalization) followed by a projection $P : \mathbb{R}^k \to \mathbb{R}^m$ (e.g., PCA), yielding projected embeddings

$$\tilde{z}_i = P(\phi(z_i)) \in \mathbb{R}^m.$$

When omitted, $\tilde{z}_i = z_i$ and $m = k$.

3.3 Trajectory Estimation

We seek a smooth embedding–outcome trajectory $g : I \to \mathbb{R}^m$ with $g \in C^2(I)$ (i.e., g is at least twice differentiable) that summarizes how embeddings vary with the ordered outcome. We parameterize g as a degree-d vector-valued B-spline:

$$g(y) = \sum_{j=1}^{p} c_j \, B_{j,d}(y)$$

where $B_{j,d}$ are B-spline basis functions of degree d and $c_j \in \mathbb{R}^m$ are coefficient vectors. For an open/unclamped B-spline construction, the total number of basis functions is $p = K + d + 1$ where K is the number of interior knots.

Estimation procedure. We form the dataset $D = \{(y_i, \tilde{z}_i)\}_{i=1}^n$ and fit the spline parameters $\{c_j\}$ to approximate $\tilde{z}_i \approx g(y_i)$. We then compute TIMEVIEW-like function characteristics of g along I (e.g., motifs, transition points) to obtain human-interpretable structure relating embeddings to outcomes.

4 Minimum Viable Prototype: EEG Biomarker Discovery

We now instantiate the generic formulation from Section 3 in the context of EEG-based biomarker discovery. We extend the framework by formalizing domain-informed desiderata as testable constraints on the embedding–outcome trajectory g, yielding quantitative criteria for evaluating candidate embeddings.

4.1 Formulation

Data. Let $D = \{(x_i, y_i)\}_{i=1}^n$ where $x_i \in X \colon \mathbb{R}^{C \times T}$ is an EEG recording with C channels and T time steps, and $y_i \in I \subset \mathbb{R}$ is an ordered outcome (e.g., cognitive score or symptom severity). Each pair (x_i, y_i) comes from the same recording/assessment instance (e.g., a task session), with y_i derived from the associated assessment.

4.2 INSTANTIATING BIOLENS

Encoder and embeddings. We use a pretrained encoder E that maps each recording to an embedding $z_i = E(x_i) \in \mathbb{R}^k$. We consider single-channel recordings sampled at 128 Hz over 10 minutes ($C = 1, T = 128 \times 10 \times 60 = 76800$) and an embedding dimension $k = 10$.

Preprocessing and projection. We apply z-scoring followed by a 1D PCA projection $P : \mathbb{R}^{10} \to \mathbb{R}$, yielding scalar embeddings $\tilde{z}_i \in \mathbb{R}$.

Trajectory estimation. We fit a cubic smoothing spline with $K = 5$ interior knots to obtain a 1D trajectory $g : I \to \mathbb{R}$ relating the ordered outcome y to the projected embedding coordinate $\tilde{z}$.

Generic and MVP-specific parameter values are summarized in Table 1.

Table 1: BioLENS parameters and MVP instantiation.

	Parameter	Description	MVP settings
Formulation	k	embedding dimension of encoder E	10
	ϕ	preprocessing operator	z-score
	P	projection operator	PCA1
	m	projection output dimension; spline vector dimension	1 (scalar)
	d	degree of spline g	3
	K	number of interior knots of spline g	5
	p	number of basis functions in spline g	9
MVP Dataset	C	number of channels in EEG signal $x_i \in X : \mathbb{R}^{C \times T}$	1
	T	number of time steps in EEG signal $x_i \in X : \mathbb{R}^{C \times T}$	76800

4.3 BIOMARKER DISCOVERY CONSTRAINTS

Desiderata as constraints. We specify clinically-motivated biomarker desiderata and express them as constraints evaluated on the fitted trajectory $g(y)$, enabling quantitative evaluation of candidate embeddings:

1. **Proximity.** Transition points occur near clinically significant boundaries.
2. **Monotonicity.** Over clinically significant ranges, the trajectory has a consistent slope direction.
3. **Reproducibility.** These properties are stable across cohorts/resamples.

We first define transition points in terms of motifs, then formalize these desiderata as testable conditions on $g(y)$.

Transition points (motif-based). Let $g : I \to \mathbb{R}$ be the embedding–outcome trajectory on an interval $I = [y_1, y_2]$. Let $T(g)$ denote the set of transition points induced by the dynamical motif composition of g (i.e., points where the local motif changes; for cubic splines these coincide with extrema and inflection points) (Kacprzyk et al., 2024).

Constraint 1: Proximity. Define the point-to-set distance $d(y, T) := \min_{t \in T} |y - t|$. We say g satisfies *proximity* to $[y_1, y_2]$ if

$$d(y_1, T(g)) \leq \tau_1 \quad \text{and} \quad d(y_2, T(g)) \leq \tau_2$$

for predefined τ_1 and τ_2.

Constraint 2: Monotonicity (motif-based). Let $C_S[g] = (s_1, \ldots, s_K)$ be the dynamical motif composition of g over $[y_1, y_2]$. Map each dynamical motif to a slope sign $\sigma(s_k) \in \{-1, 0, +1\}$ by $\sigma(s) = \text{sign}(g')$ on the interval corresponding to s. Let $N_+ := |\{k : \sigma(s_k) = +1\}|$ and $N_- := |\{k : \sigma(s_k) = -1\}|$. We say g satisfies *monotonicity* on $[y_1, y_2]$ if

$$\frac{\max(N_+, N_-)}{K} \geq \rho$$

for a predefined $\rho \in [0, 1]$.

Constraint 3: Reproducibility. Let $C(D)$ be any boolean-valued condition evaluated on dataset D. For bootstrap resamples $D^{(1)}, \ldots, D^{(B)}$, define

$$\widehat{R} := \frac{1}{B} \sum_{b=1}^{B} \mathbb{1}\{C(D^{(b)})\}.$$

We say C satisfies *reproducibility* if $\widehat{R} \geq \alpha$, i.e., the condition holds in at least an α fraction of bootstrap resamples. In practice, we apply reproducibility to the proximity and monotonicity conditions.

Fit adequacy (precondition). In addition to satisfying constraints 1-3, the fitted $g(y)$ should be sufficiently accurate such that derivative- and motif-based characteristics are meaningful. We note two potential checks:

- **Comparison to null distribution.** Compute mean squared error (MSE) on a holdout set. Generate a null distribution by repeatedly shuffling y, refitting $g(y)$, and computing holdout MSE. Compare the true holdout MSE to this null distribution (e.g., via a one-sided permutation p-value) (Ojala & Garriga, 2010). A statistically significant improvement over the null indicates that the observed fit is unlikely to arise from chance alignment between embeddings and labels, and therefore supports the claim that the embedding contains information about y.
- **Improvement relative to baseline.** Compute holdout MSE of $g(y)$ and compare to a baseline predictor (e.g., training set mean prediction, linear regression). An improvement in MSE indicates that the embedding-label relationship is not only non-random, but also that the spline captures structure beyond what is explained by a simpler relationship; in particular, improvement over a linear baseline suggests the presence of meaningful nonlinear (yet smooth) dependence of the embedding on y.

An example of a viable and non-viable biomarker assessed per these constraints can be found in Figure 2.

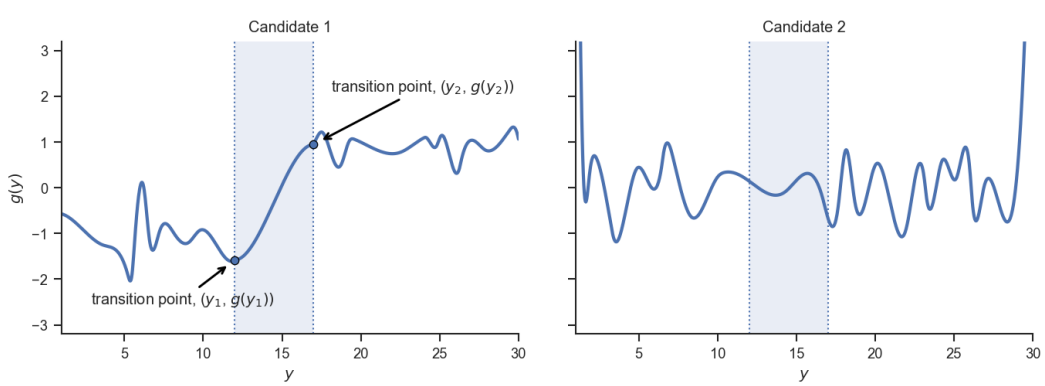

Figure 2: Viable vs. non-viable biomarker examples. Candidate 1 (left) meets proximity and monotonicity in $[y_1, y_2]$; Candidate 2 (right) does not.

4.4 TARGETED REPRESENTATION LEARNING

A further advantage of BioLENS is the flexibility it affords in evaluating representation learning frameworks. Rather than treating representation quality as synonymous with downstream task performance or relying on latent space diagnostics whose practical meaning is unclear, BioLENS evaluates embeddings using mechanistic criteria tied to specific clinical endpoints. Concretely, we score candidate representations by asking whether the embedding behaves like a plausible biomarker: does it show a coherent directional relationship over relevant ranges, remain stable under perturbations such as resampling or noise, and exhibit signal that is distinguishable from chance? Aggregating these scores yields an evaluation matrix such as that in Table 2. This reframes model selection from "which model fine-tunes best" to "which model learns embeddings that capture outcome-relevant structure," while still remaining compatible with standard downstream evaluation.

Table 2: Example of targeted representation learning framework.

Model	Proximity	Monotonicity	Reproducibility	Fit	Selected
CNN, masked	✓	✗	✓	✗	✗
CNN, contrastive	✗	✗	✗	✓	✗
Transformer, masked	✓	✓	✓	✓	✓
Transformer, contrastive	✗	✓	✓	✗	✗
RWKV, masked	✓	✗	✗	✓	✗
RWKV, contrastive	✓	✓	✗	✗	✗

5 RELATED WORKS

Here, we explain how our work intersects with related areas of ML and EEG-based biomarker discovery.

Foundation model embedding interpretability. Most embedding interpretability work for foundation models is post hoc. For example, linear probes and diagnostic classifiers test what is decodable from frozen representations (Kim et al., 2018), while concept-based methods quantify the influence of human-defined directions on predictions (Koh et al., 2020). A complementary line of work notes that interpretability can be a coordinate artifact; for example, axis-aligned semantics may disappear under rotations even though the model's behavior is unchanged (Bau et al., 2017). In contrast, BioLENS is not primarily a probe: it treats embeddings as substrates for constraint-driven, trajectory-level explanations by explicitly modeling a smooth embedding–outcome curve and evaluating it for domain-informed properties, making interpretability a target rather than a retrospective diagnostic.

EEG-based biomarker discovery. EEG-based biomarker discovery has traditionally focused on interpretable feature families (bandpower, event-related potentials, connectivity) (Biasiucci et al., 2019), but these are often fragile to preprocessing, cohort heterogeneity, and other confounding factors (Soares de Aguiar Neto & Garcia Rosa, 2019). Recent work explores latent spectral descriptors that track symptom dimensions (Hacker et al., 2025), while self-supervised EEG representations often remain weakly linked to clinically actionable biomarkers despite effective transferability (Kostas et al., 2021; Chau et al., 2025). BioLENS aims to close this gap by converting pretrained embeddings into explicit, testable biomarker hypotheses informed by domain expectations rather than relying only on downstream task performance.

Implicit representation learning objectives. Self-supervised learning makes clear that objectives define latent geometry: contrastive instance discrimination (Chen et al., 2020; He et al., 2020) and masked reconstruction (He et al., 2022) yield different invariances, while time series SSL further injects temporal assumptions via neighborhood or multi-scale objectives (Tonekaboni et al., 2021; Yue et al., 2022). Separately, shape-constrained modeling shows how domain priors can be enforced directly through differentiable architectures/regularizers (You et al., 2017). BioLENS is novel in combining these ideas: it operationalizes interpretability as an implicit learning objective on the

representation itself (via constraints on the embedding–outcome trajectory), creating a pathway from generic representation learning to embeddings whose geometry is deliberately aligned with plausible outcome progression.

6 Limitations and Future Work

BioLENS makes interpretability concrete by fitting a smooth embedding–outcome trajectory g, aligning with TIMEVIEW's premise that humans reason about temporal structure through trends and their properties. While this yields a clean interface for derivative- and motif-based statements, several limitations remain and motivate future work.

Score fidelity and robustness. The monotonicity score outlined in the MVP is intentionally simple for demonstrative purposes, counting slope sign prevalence across motif segments and thresholding by ρ. It ignores segment durations and can be unstable when motif composition is sparse. It also discards second-order information despite the natural role of extrema and inflection points in the motif formalism. Future work should incorporate duration weighting, uncertainty in low-motif regimes, and second-order-aware constraints.

Thresholding and domain knowledge. BioLENS relies on thresholds τ_1, τ_2, ρ, and α for proximity, monotonicity, and reproducibility constraints. These may be brittle across cohorts and outcomes. By incorporating domain knowledge as priors or explicit utility objectives, the selection of these thresholds could be made more principled and portable.

Projection and scalarization. The MVP projects embeddings to scalar values (PCA1) prior to spline fitting. This can discard relevant structure and make conclusions sensitive to an unsupervised axis (Shlens, 2014). A key next step is to learn the projection in an outcome-aware, constraint-aware manner, or to avoid scalarization via a low-dimensional vector-valued trajectory.

Dimensionality and tractable interpolation. Although the generic formulation supports vector-valued trajectories, scaling interpolation and interpretation to higher-dimensional latent spaces quickly becomes challenging, requiring increasing amounts of data to resolve local behavior and decreasing the extent of interpretability. Low-rank spline coefficients and related approaches could preserve more information while keeping the trend abstraction usable.

7 Conclusion

We introduced BioLENS, a TIMEVIEW-inspired framework for interpreting embeddings in ordered-outcome settings by modeling representations as a smooth embedding–outcome trajectory g and extracting derivative- and motif-based structure to produce human-interpretable summaries. In our EEG-based MVP, we instantiated g using standardized embeddings, a low-dimensional projection, and a smoothing spline. We then explored how the resulting trajectory behavior could be used to formulate mathematical characteristics of viable biomarkers: proximity (whether key transitions occur near clinically meaningful thresholds), monotonicity (whether behavior is consistent over outcome ranges of interest), reproducibility (whether these properties persist under bootstrap resampling), and fit adequacy checks to guard against over-interpreting noise or poorly fit trajectories.

Our MVP is best understood as one instance of the broader principle that, by leveraging a BioLENS-like approach, domain-informed desiderata can be formalized as explicit, testable constraints on the relationship between an ordered outcome and candidate embedding. Beyond explanation, BioLENS offers a practical route to selecting, and ultimately learning, representations whose latent geometry is not only predictive but also aligned with plausible outcome structure, enabling a clearer bridge from foundation model embeddings to testable hypotheses.

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